

Multiple Choice

1. **Answer:** A

Options: A) real; B) imaginary; C) degenerate; D) linear

Note: The eigenvalues of a Hermitian operator are always real because Hermitian operators, which represent observable quantities in quantum mechanics, have the property that their matrix elements are equal to their complex conjugates, leading to real eigenvalues when solving the corresponding eigenvalue equations.

2. **Answer:** A

Options: A) decreased by a factor of 9; B) decreased by a factor of 49; C) decreased by a factor of 81; D) increased by a factor of 9

Note: The emission spectrum of the Li^{++} ion is identical to that of hydrogen because both have one electron; however, since Li^{++} has a higher nuclear charge ($Z = 3$) compared to hydrogen ($Z = 1$), the wavelengths of emitted photons are decreased by a factor of 9 (which is the square of the ratio of the atomic numbers, $3/1$).

3. **Answer:** D

Options: A) $(1/2) \text{ k T}$; B) kT ; C) $(3/2) \text{ k T}$; D) 3 kT

Note: The average total energy of a three-dimensional harmonic oscillator in thermal equilibrium at temperature T is given by the equipartition theorem, which states that each quadratic degree of freedom contributes an average energy of $(1/2)\text{kT}$, and since there are three spatial dimensions, the total average energy is $3(1/2)\text{kT} = 3 \text{ kT}$.

4. **Answer:** B

Options: A) Constant temperature; B) Constant volume; C) Constant pressure; D) Adiabatic

Note: In a constant volume process, no work is done on or by the gas, so the change in internal energy is equal to the heat added to the system, according to the first law of thermodynamics.

5. **Answer:** D

Options: A) 44 Hz; B) 196 Hz; C) 262 Hz; D) 393 Hz

Note: The next higher harmonic for an organ pipe closed at one end and open at the other is the third harmonic, which has a frequency that is three times the fundamental frequency, resulting in $3 \times 131 \text{ Hz} = 393 \text{ Hz}$.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because for a given angular momentum quantum number

$l = 2$, the magnetic quantum number m_l can take on values ranging from $-l$ to $+l$, resulting in a total of 5 possible values: -2 , -1 , 0 , $+1$, and $+2$.

7. **Answer:** True

Options: A) True; B) False

Note: Electromagnetic radiation emitted from a nucleus is most likely to be in the form of gamma rays because gamma radiation is produced during radioactive decay processes when the nucleus transitions from a higher energy state to a lower energy state, releasing excess energy in the form of high-energy photons.

8. **Answer:** True

Options: A) True; B) False

Note: A dye laser is the most suitable type of laser for spectroscopy across a range of visible wavelengths because it can be tuned to emit a wide variety of wavelengths by changing the dye used, allowing for precise measurements in various spectroscopic applications.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the time dilation effect in the lab frame means that the particle will travel a shorter distance than 360 m before decaying, as its proper decay time of 2.0 ms is longer when observed in the lab frame due to its relativistic speed of $0.60c$.

10. **Answer:** True

Options: A) True; B) False

Note: In a reversible thermodynamic process, the system and its surroundings can be returned to their original states without any net change in entropy, as all processes occur infinitely slowly and without dissipative effects.

Long Form Response

11. **Answer:** A beam of electrons can diffract on a crystal surface when certain conditions are met, primarily when the electron wavelength is comparable to the lattice spacing of the crystal. For a lattice spacing of 0.4 nm, the de Broglie wavelength of the electrons must be on the order of this length to observe distinct diffraction patterns. The relationship between the kinetic energy of the electrons and the scattering pattern is governed by the de Broglie wavelength, which is inversely proportional to the square root of the kinetic energy. Specifically, electrons with a kinetic energy of approximately 10 eV yield a de Broglie wavelength of about 0.12 nm, allowing them to interact effectively with the crystal lattice and produce observable diffraction. Thus, achieving a kinetic energy around 10 eV is necessary for the diffraction phenomenon to occur under these conditions.

Note: correct because it identifies that for electron diffraction to occur, the de Broglie wavelength must match the crystal lattice spacing, and it explains that the kinetic

energy of the electrons is inversely related to their wavelength, thus determining the conditions necessary for observing diffraction patterns.

12. **Answer:** De Broglie hypothesized a fundamental relationship between the linear momentum (p) of a free massive particle and its wavelength (λ) through the equation $\lambda = h/p$, where h is Planck's constant. This relationship implies that every particle exhibits wave-like behavior, with its wavelength inversely proportional to its momentum. Planck's constant serves as a crucial proportionality factor, linking the macroscopic world of classical mechanics with the microscopic realm of quantum mechanics. In essence, as the momentum of a particle increases, its wavelength decreases, highlighting the dual wave-particle nature of matter and the significance of h in quantifying this behavior. This groundbreaking insight laid the foundation for modern quantum theory, illustrating the intrinsic connection between physical properties that were once thought to be distinct.

Note: correct because it accurately describes De Broglie's hypothesis that associates a particle's wavelength with its momentum through the equation $\lambda = h/p$, illustrating how Planck's constant serves as a bridge between classical and quantum mechanics by establishing the wave-particle duality of matter.